

ME 333
MECHANICS OF MATERIALS

Lecture 9: Concept of Stress - Tutorial Problems

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Tutorial Problems

[Quiz-1 Problem] Mr. Gagan Vihari is a senior aircraft technician at Boeing who has been assigned to measure the two dimensional state of strain at various locations on an aircraft wing. At one critical location O on the wing (thin plate), Gagan used a strain gauge rosette having arms A, B and C with counterclockwise angles $\angle AOB = 30^\circ$ and $\angle BOC = 60^\circ$. The strain gauge readings in arms A and C were e_a and $-3e_a$, respectively. However, the strain gauge in arm B was damaged and could not measure the strain. With years of experience and from data at other locations on the wing, Gagan inferred that the maximum in-plane shear strain (ϵ_{xy}^{max}) permissible at O was $2e_a \sqrt{7/3}$. Help Gagan determine the strain reading in arm B and the state of strain at the point O . [Note: Do not use graphical approach]

Answer: $\epsilon_{xx} = e_a$, $\epsilon_{yy} = -3e_a$, $\epsilon_{xy} = \pm \frac{4}{\sqrt{3}}e_a$, $\epsilon_B = \pm 2e_a$

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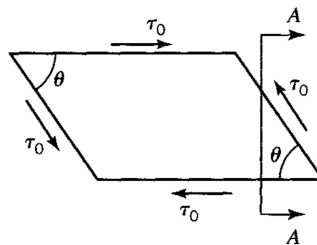
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[Problem 2.2] A thin flat plate is subjected to uniform shear stress. Find the normal and shear stress acting on the cutting plane A-A.



Answer: $\sigma_{xx} = -2\tau_0 \cot \theta$; $\sigma_{xy} = \tau_0$

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[Example 1] Given the state of stress (in MPa):

$$\sigma = \begin{bmatrix} 15 & 10 & -10 \\ 10 & 10 & 0 \\ -10 & 0 & 40 \end{bmatrix}$$

Determine the normal and shear stress on a surface intersecting the point and parallel to the plane given by: $2x - y + 3z = 9$.

Answer: $\sigma_{NN} = 19.286 \text{ MPa}$; $\sigma_{NS} = 18.89 \text{ MPa}$

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[Example 2] Given the state of stress (in MPa):

$$\sigma = \begin{bmatrix} \sigma & 2 & 1 \\ 2 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$$

Determine σ such that there is atleast one plane passing through the point in such a way that the resultant stress on that plane is zero.

Answer : $\sigma = 2$ MPa

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[Problem 2.4] Stress components at a point in a plate are:

$$\sigma = \begin{bmatrix} -100 & 0 & -80 \\ 0 & 20 & 0 \\ -80 & 0 & 20 \end{bmatrix}$$

a) Determine principal stresses.

Answer : $\sigma_1 = 60$ MPa, $\sigma_2 = 20$ MPa and $\sigma_3 = -140$ MPa.

Tutorial Problems

[Problem 2.4]

b) Determine Octahedral stresses

Answer : $\sigma_{\text{Oct}} = -20$ MPa; $\tau_{\text{Oct}} = 86.41$ MPa.

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[Problem 2.4]

c) Determine max. shear stress

Answer : 100 MPa.

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[Problem 2.4]

d) Determine direction cosines of the normal to the plane on which the maximum shear stress acts.

Answer : $(\pm 1/\sqrt{2}, 0, \pm 1/\sqrt{2})$ with respect to the principal axis.

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[Problem 2.4]

e) Determine the invariants.

Answer : $I_1 = -60$, $I_2 = -10000$ and $I_3 = -168000$.

Tutorial Problems

[Problem 2.4]

f) Determine the stresses on a plane oriented at an angle of 45° to the $x - y$ plane when rotated about z axis.

Answer :

$$\sigma = \begin{bmatrix} -40 & 60 & -56.569 \\ 60 & -40 & 56.569 \\ -56.569 & 56.569 & 20 \end{bmatrix}$$

Tutorial Problems

[Problem 2.4]

g) Determine invariants in the transformed plane.

Answer : $I'_1 = -60$, $I'_2 = -10000$ and $I'_3 = -168000$.

Tutorial Problems

[Example 3] Given the stress field:

$$\sigma_{xx} = 3xy^2z + 2x; \quad \sigma_{yy} = 5xyz + 3y; \quad \sigma_{zz} = x^2y + y^2z$$

$$\tau_{xy} = 0; \quad \tau_{yz} = \tau_{zx} = 3xy^2z + 2xy$$

Determine whether the components of stress satisfy equilibrium equations or not at a point $(1, -1, 2)$. If not then determine the suitable body forces.

Answer : $B = -11\hat{i} - 16\hat{j} + 5\hat{k}$

Reading Assignment

Chapter 2: Section 2.1 - 2.5